

SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22



Programme: B. Tech CSE Cyber Security

Year: II Semester: IV

Subject: Introduction to Cryptography

Date: 09 June 2022

Marks: 100

Time: 10.00 am to 1.00 pm

Durations: 3 (Hrs)

No. of Pages: 04.

Re-Examination (2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) Assume Suitable data if necessary.

Q1		Answer the following questions	[20]
	a.	Suppose that the Affine Cipher with the encryption function $e(x) = 9x + 3 \pmod{26}$ is used to encrypt a plaintext. Find the decryption function $d(y) = cy + d \pmod{26}$	
	b.	If the Affine Cipher in the previous problem (Q1a) yields the ciphertext as SABN, find the corresponding plaintext.	
	c.	The following text has been obtained by first encoding the plaintext by using Vigenère Cipher with the key "BOND" and then encoding the result again by Caesar Cipher. Decrypt the text: GYYVWRDJHZFY	
	d.	Decrypt the following cipher text obtained from the Autokey Cipher by using the key $k = 19$: ALVVXHGLTY	
	e.	By using unicity distance formula $n_0 = \frac{4 \log_2 K }{3 \log_2 P }$, find the minimum number of ciphertext length usually needed to uniquely decrypt for a Substitution Cipher. You should select your answer as the smallest integer higher than your answer from the formula. For example, if your formula yields 3.2, then the answer should be 4.	
Q2		Consider the following: Suppose that a player attempts five free throws in a basketball game. Let X be the number of free throws made out of five attempts. So the possible outcomes are $\{0, 1, 2, 3, 4, 5\}$. Suppose that X has the following probability distribution:	[20]

		<table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>p[x]</td><td>0.05</td><td>0.10</td><td>0.18</td><td>0.26</td><td>0.24</td><td>0.17</td></tr><tr><td>code f(x)</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> Find the Huffman code f(0), f(1), f(2), f(3), f(4), f(5) and Compute the average codeword length. Also compute Shannon entropy H(X).	X	0	1	2	3	4	5	p[x]	0.05	0.10	0.18	0.26	0.24	0.17	code f(x)							
X	0	1	2	3	4	5																		
p[x]	0.05	0.10	0.18	0.26	0.24	0.17																		
code f(x)																								
Q3	A	Explain Data Encryption standard (DES) in detail.	[10]																					
	B	Alice wants to use RSA algorithm for encryption and decryption. What will be her private key if following values are used? P=13; q=17; public key, e=23, private key, d=? If the message is 5, what will be the encrypted value?	[10]																					
Q4	A	Let S ₁ and S ₂ be the standard Vigenère and Permutation ciphers, with block lengths 3 and 5 respectively. Consider the product cipher S ₁ X S ₂ . Consider the keycode k ₁ = cat in Vigenère Cipher, and the key k ₂ in Permutation Cipher given by $\pi = \begin{array}{ c c c c c } \hline 1 & 2 & 3 & 4 & 5 \\ \hline 4 & 5 & 2 & 1 & 3 \\ \hline \end{array}$ Construct π^{-1} and find the decryption d(k ₁ ,k ₂)(GSAJQAQHTEARBLF) in S ₁ X S ₂ . Find your plaintext with spaces.	[04]																					
	B	Consider the cryptosystem in which P = {a, b, c}, K = {k ₁ , k ₂ , k ₃ } and C = {1, 2, 3, 4} with p[a] = 1/2, p[b] = 1/3, p[c] = 1/6 and the keys are chosen equiprobably, that is, p[k ₁] = p[k ₂] = p[k ₃] = 1/3. The encryption matrix is given as follows: <table><tr><td></td><td>a</td><td>b</td><td>c</td></tr><tr><td>k₁</td><td>1</td><td>2</td><td>3</td></tr><tr><td>k₂</td><td>2</td><td>3</td><td>4</td></tr><tr><td>k₃</td><td>3</td><td>4</td><td>1</td></tr></table> - Find p[1] - Find p[2] - Find the conditional probability p[3 b]. - By using Bayes' theorem or directly, find the conditional probability p[b 3]. - Find the joint probability p[b, 3] - By using the formula $H(X) = -\sum p[x] \log_2 p[x]$, compute H(P) - Compute H(K) - Compute H(C)		a	b	c	k ₁	1	2	3	k ₂	2	3	4	k ₃	3	4	1	[16]					
	a	b	c																					
k ₁	1	2	3																					
k ₂	2	3	4																					
k ₃	3	4	1																					
Q5	A	Explain computational security, provable security and unconditional security.	[10]																					
	B	Let (P, C, K, E, D) be a cryptosystem such that P = {m, a, p, s}, C = {S, A, M, E} and K = {k ₁ , k ₂ }. Encryption is performed according to the following table: <table><tr><td></td><td>m</td><td>a</td><td>p</td><td>s</td></tr><tr><td>k₁</td><td>S</td><td>A</td><td>M</td><td>E</td></tr><tr><td>k₂</td><td>M</td><td>E</td><td>S</td><td>A</td></tr></table>		m	a	p	s	k ₁	S	A	M	E	k ₂	M	E	S	A	[10]						
	m	a	p	s																				
k ₁	S	A	M	E																				
k ₂	M	E	S	A																				

	The probability distribution of plaintext characters is given by: <table><tr><td>plaintext</td><td>m</td><td>a</td><td>p</td><td>s</td></tr><tr><td>probability</td><td>1/2</td><td>1/4</td><td>1/8</td><td>1/8</td></tr></table> Assume that $p[k_1] = 1/4$ and $p[k_2] = 3/4$. Find the following 1. $p[S]$ 2. $p[S m]$ 3. $p[m S]$ 4. Does the system exhibit perfect secrecy? Justify your answer	plaintext	m	a	p	s	probability	1/2	1/4	1/8	1/8																																																																																																																														
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probability	1/2	1/4	1/8	1/8																																																																																																																																					
Q6	A Consider the following S-box π_S for a simple SPN: <table><tr><td>input</td><td>000</td><td>001</td><td>010</td><td>011</td><td>100</td><td>101</td><td>110</td><td>111</td></tr><tr><td>output</td><td>110</td><td>101</td><td>001</td><td>000</td><td>011</td><td>010</td><td>111</td><td>100</td></tr></table> Complete the following table: <table><tr><td>X_1</td><td>X_2</td><td>X_3</td><td>Y_1</td><td>Y_2</td><td>Y_3</td><td>$X_1 \oplus X_3$</td><td>$Y_2 \oplus Y_3$</td></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr></table> Compute the following 1. $N_L(5, 3)$ 2. bias $\epsilon(5, 3)$ 3. Can the pair $(5, 3)$ be used to construct linear approximation for the given SPN. 4. $R_p(5, 6)$ 5. $N_D(5, 6)$ for the S-box by filling out the following table with $x' = 101$ <table><tr><td>x</td><td>x^*</td><td>y</td><td>y^*</td><td>y'</td></tr><tr><td>000</td><td></td><td>110</td><td></td><td></td></tr><tr><td>001</td><td></td><td>101</td><td></td><td></td></tr><tr><td>010</td><td></td><td>001</td><td></td><td></td></tr><tr><td>011</td><td></td><td>000</td><td></td><td></td></tr><tr><td>100</td><td></td><td>011</td><td></td><td></td></tr><tr><td>101</td><td></td><td>010</td><td></td><td></td></tr><tr><td>110</td><td></td><td>111</td><td></td><td></td></tr><tr><td>111</td><td></td><td>100</td><td></td><td></td></tr></table>	input	000	001	010	011	100	101	110	111	output	110	101	001	000	011	010	111	100	X_1	X_2	X_3	Y_1	Y_2	Y_3	$X_1 \oplus X_3$	$Y_2 \oplus Y_3$	0	0	0						0	0	1						0	1	0						0	1	1						1	0	0						1	0	1						1	1	0						1	1	1						x	x^*	y	y^*	y'	000		110			001		101			010		001			011		000			100		011			101		010			110		111			111		100			[10]
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B	Consider the Baby Horst Cipher with two rounds where $L^i = R^{i-1}$ and $R^i = L^{i-1} \oplus f(R^{i-1}, k_i)$ and f is given by the formulas: $f(x, k) = (2k_1)^x \bmod 16$ for the first																																																																																																																																								

round, whereas $f(x, k) = (4k_2)^x \bmod 16$ for the second round (here both x and k are considered as hexadecimal numbers, for example, $1100 = C = 12$). Consider the round keys $k_1 = A$ and $k_2 = 3$.

1. If the plaintext is $P = 0010\ 0001 = 21$, find L^1R^1 in hexadecimal notation.
2. If the ciphertext is $C = 0010\ 1010 = 2A$, find L^1R^1 in hexadecimal notation by using $R^{i-1} = L^i$ and $L^{i-1} = R^i \oplus f(L^i, k_i)$.

Alphabet Codings

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Inverses in Z_{26}

a	1	3	5	7	11	17	25
a^{-1}	1	9	21	15	19	23	25